Project Proposal

Secure Personal Finance Tracker

Sahibenoor Singh Sidhu

3rd year ACS

University of Winnipeg

Introduction

The secure personal finance tracker is a simple web application that helps users to record, manage and analyse their finances. The goal of my first ever CS project is to build a complete and beginner friendly full-stack application using technologies such as Node, Express, Postgres, HTML, CSS and JavaScript.

This project is designed as my first solo software project. I may not be able to build a advanced finance app but I will try my best to build a clean, secure and working version 1.

The Application will let users to create an account, log in, add transactions, view history, filter records and see organized dashboard. I will apply my knowledge from programming, database systems, internet programming, requirements analysis, software design, and computer security courses.

Problem Statement

Many people these days spend their money without a proper track of it and where it goes. In earlier times, people were so cautious of their spendings and people used to track everything on paper like how much they spend on food, transport, housing, entertainment, or other categories each month. People used to keep receipts and records of their transactions. That was a lot of hard work and really time consuming. Nowadays, people neither have spare time nor the motivation to track their expenses.

So, if there are people who are willing to do the same like earlier times however want quick and easy alternative to track their finances, I am creating a simple and secure personal finance tracker where a user can manually record income and expenses and quickly understand their monthly financial activity.

Project Objectives

The Core objectives of this project are:

1. To build a secure login and registration system.
2. To allow users to add, edit, delete, and view transactions.
3. To store user and transaction data safely in a PostgreSQL database.
4. To make sure each user can only access their own financial data.
5. To show monthly income, monthly expenses, and monthly net balance.

6. To display a simple chart showing spending by category.
7. To create a clean project structure with documentation and testing.
8. To allow users to send a monthly spending summary to their registered email address.

Target Users

The main target users are students, entry-level employees, beginner budget conscious people who really want a simplified way of tracking money. The users who may not (at this time) need advanced features like bank integrations, investment planning, advanced automated reports or forecasting.

The application is mainly for users who want to manually record their transactions and review monthly spending in clear and simple way.

Main Features

Version 1 features:

Authentication:

Users can register with an email and password. They can log in and log out. Passwords will be hashed before being stored, so plain text passwords will not be saved. Sessions will be used to keep users logged in.

Transactions:

Users can add a transaction with amount, type, category, date and optional note. The transaction type will be either income or expense. Users will also be able to edit and delete their own transactions.

Categories:

The application will use a fixed list of categories: Food, Transport, Housing, Entertainment, Health, Salary, and Other. Custom category creation will be the scope for version 2.

Transaction List:

Users can view all their transactions in a table. The table will show transaction details such as amount, type, category, date, and note.

Filters:

Users will be able to filter transactions by month/year, by transaction type, such as income or expense, by category such as food.

Dashboard:

The dashboard will show total income for the current month, total expenses for the current month, and net balance for the current month. It will also show one simple bar chart or pie chart for spending by category.

Monthly Email Report:

Users will be able to click a button to send a monthly spending summary to their registered email address. The email will include the selected month, total income, total expenses, net balance, and spending by category. This feature will use an external Email API from the backend.

Technologies Used

Node.js to run JavaScript on the server side.

Express.js to create the backend routes and handle requests from the browser.

PostgreSQL as the relational database for storing users and transactions.

HTML, CSS, and JavaScript for the frontend interface.

Bootstrap to make forms, buttons, tables, and pages look clean without spending too much time on design.

bcrypt to hash user passwords.

express-session to manage user login sessions.

Chart.js to create the spending-by-category chart.

Email API such as Resend to send monthly spending summary emails.

dotenv to store API keys and environment variables securely.

Git and GitHub for version control and project storage.

Database Plan

The database will be designed in PostgreSQL. The main tables will be:

users table

Table to store user account information.

Possible columns:

- user_id
- email
- password_hash
- created_at

email_logs table

This optional table will store records of monthly report emails sent by the system.

Possible columns:

• email_log_id

• user_id

• recipient_email

• email_type

• status

• sent_at

Each email log will belong to one user. This table will help track when a monthly report email was sent successfully or failed.

transactions table

This table to store income and expense records.

Possible columns:

- transaction_id
- user_id
- amount
- type
- category
- transaction_date

- note
- created_at

Each transaction will belong to one user. The user_id in the transactions table will be a foreign key connected to the users table. This is important because each user must only see and manage their own transactions.

For Version 1, categories may be stored as a fixed text value instead of creating a separate custom categories system.

Project Scope

The scope of Version 1 is intentionally limited. The project focuses on building the core features properly instead of adding too many extra features.

Included in Version 1

- User registration
- Login and logout
- Password hashing
- Session management
- Add transaction
- Edit transaction
- Delete transaction
- View transactions
- Fixed categories
- Dashboard totals
- Spending-by-category chart
- Send monthly spending summary by email
- Filters
- Basic security checks
- Testing and documentation

Expected Challenges

One expected challenge is learning and becoming more comfortable with Express.js and backend JavaScript. My main programming background is Java, so I will need to carefully understand asynchronous JavaScript, routes, middleware, and database queries.

Another challenge will be connecting PostgreSQL with Express and writing correct SQL queries. I will need to make sure the database design is clean before writing too much code.

Security will also be an important challenge. Passwords must be hashed, sessions must be handled properly, and users must not be able to access another user's transactions.

One challenge will be integrating an external Email API. I will need to learn how to call the API from the Express backend, store the API key safely using environment variables, and make sure the email feature does not expose sensitive user information.

Another challenge may be managing time and avoiding extra features. Since this is a first serious project, it will be important to follow the locked Version 1 scope and not add unnecessary features.

Expected Outcome

By the end of project, I expect to have a working secure personal finance tracker. A user should be able to register, log in and logout, add income and expenses, view transactions, filter them, see a monthly dashboard with totals and a chart, and send a monthly spending summary to their registered email address.

The final project should include source code, database schema, documentation, test results, screenshots, README file, final report, and presentation.

Conclusion

The Secure Personal Finance Tracker is a realistic first full-stack project. It is not meant to be a complex finance platform. However, it focuses on the most important parts of a beginner project: secure authentication, database design, CRUD operations, user-specific data, dashboard summaries, testing, and documentation.

This project would help me connect what I have learned in university with a practical application. By keeping the scope simple and controlled, I can finish a working Version 1 properly and later add more advanced features as Version 2 ideas.